

Harmless and Helpful Tradeoff - Physical AI Actions

Safety (Harmlessness)

Safe

Safe · Inefficient

안전하나 비효율적

Low operational value

Efficient · Safe

효율적이고 안전한

Desired region

Unsafe

Unsafe · Inefficient

위험하고 비효율적

Worst on both axes

Efficient · Unsafe

효율적이나 위험한

Highest physical-harm risk

Inefficient

Efficient

Task Efficiency (↓ time, ↓ energy)

Efficiency is defined given task completion (else minimizing time/energy degenerates to inaction).